Chapter 11. Bayesian Inference

Prior and posterior, Bayesian inference, Bayes and minimax estimators

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11.1 Prior and posterior distributions

Bayesian inference describes uncertainty about $\theta \in \Omega$ before the data by a **prior density** $\pi(\theta)$ (whose existence and uniqueness is the source of a long controversy; here it is given). We write $f(x \mid \theta)$ for the model density.

Theorem 11.1 (Bayes' theorem)

If $X_1, \dots, X_n$ is a sample from $f(x \mid \theta)$ and θ has prior density $\pi(\theta)$, then given $\mathbf{X} = \mathbf{x}$ the density of θ is

$$\pi(\theta \mid \mathbf{x}) \propto L(\theta; \mathbf{x})\pi(\theta), \qquad L(\theta; \mathbf{x}) = \prod_{i=1}^{n} f(x_i \mid \theta).$$

Proof. $\pi(\theta \mid \mathbf{x}) = f(\mathbf{x} \mid \theta)\pi(\theta)/f_{\mathbf{X}}(\mathbf{x})$ with $f_{\mathbf{X}}(\mathbf{x}) = \int_\Omega f(\mathbf{x} \mid \theta)\pi(\theta)\, d\theta$ free of θ. ∎

$\pi(\theta \mid \mathbf{x})$ is the **posterior density**: posterior $\propto$ likelihood $\times$ prior. An **improper** prior ($\int \pi = \infty$) may be used if the posterior is proper.

Example 11.1 (Binomial–beta)

Bernoulli(θ) data with prior Beta(α, β): with $t = \sum x_i$,

$$\pi(\theta \mid \mathbf{x}) \propto \theta^t (1-\theta)^{n-t} \theta^{\alpha-1} (1-\theta)^{\beta-1},$$

so $\theta \mid \mathbf{x} \sim \text{Beta}(t+\alpha, n-t+\beta)$.

Example 11.2 (Poisson–gamma)

Poisson(θ) data with prior Gamma(α, β): $\pi(\theta \mid \mathbf{x}) \propto e^{-n\theta}\theta^t \theta^{\alpha-1} e^{-\theta/\beta}$, so $\theta \mid \mathbf{x} \sim \text{Gamma}(t+\alpha, (n+\beta^{-1})^{-1})$.

Example 11.3 (Exponential–gamma)

Exp(θ) data, rate $\lambda = 1/\theta \sim \text{Gamma}(\alpha, \beta)$: $\pi(\lambda \mid \mathbf{x}) \propto \lambda^n e^{-\lambda t} \lambda^{\alpha-1} e^{-\lambda/\beta}$, so $\lambda \mid \mathbf{x} \sim \text{Gamma}(n+\alpha, (t+\beta^{-1})^{-1})$; $\pi(\theta \mid \mathbf{x}) = \pi(\lambda \mid \mathbf{x})\theta^{-2}$.

Example 11.4 (Normal mean, known variance)

$N(\theta, \sigma^2)$ data, σ^2 known, prior $N(\mu_0, \sigma_0^2)$:

$$\pi(\theta \mid \mathbf{x}) \propto \exp\Big\{ -\frac{n}{2\sigma^2}(\theta - \bar{x})^2 - \frac{1}{2\sigma_0^2}(\theta - \mu_0)^2 \Big\}.$$

Completing the square, $\theta \mid \mathbf{x} \sim N(\mu(\mathbf{x}), \sigma^2(\mathbf{x}))$ with

$$\mu(\mathbf{x}) = \frac{n\sigma^{-2}\bar{x} + \sigma_0^{-2}\mu_0}{n\sigma^{-2} + \sigma_0^{-2}}, \qquad \sigma^2(\mathbf{x}) = (n\sigma^{-2} + \sigma_0^{-2})^{-1}.$$

Posterior precision = data precision + prior precision.

A prior whose posterior stays in its family is **conjugate**.

Theorem 11.2 (Normal model, conjugate prior)

Let $X_1, \dots, X_n$ be $N(\mu, \sigma^2)$, and let the prior of $\theta = (\mu, \sigma^2)$ be

$$\mu \mid \sigma^2 \sim N(\mu_0, \sigma^2/\kappa_0), \qquad \nu_0\sigma_0^2/\sigma^2 \sim \chi^2(\nu_0).$$

Put $\kappa_n = n + \kappa_0$, $\nu_n = n + \nu_0$, $s^2 = \sum(x_i - \bar{x})^2/(n-1)$, and

$$\mu_n(\mathbf{x}) = \frac{n\bar{x} + \kappa_0\mu_0}{n + \kappa_0}, \qquad \nu_n\sigma_n^2(\mathbf{x}) = \nu_0\sigma_0^2 + (n-1)s^2 + (\kappa_0^{-1} + n^{-1})^{-1}(\bar{x} - \mu_0)^2.$$

Then

(a) $\mu \mid \sigma^2, \mathbf{x} \sim N(\mu_n(\mathbf{x}), \sigma^2/\kappa_n)$ and $\nu_n\sigma_n^2(\mathbf{x})/\sigma^2 \mid \mathbf{x} \sim \chi^2(\nu_n)$;

(b) $\dfrac{\mu - \mu_n(\mathbf{x})}{\sigma_n(\mathbf{x})/\sqrt{\kappa_n}} \,\Big|\, \mathbf{x} \sim t(\nu_n)$.

Proof. (a) With $\tau = \sigma^{-2}$,

$$L \propto \tau^{n/2} \exp\Big\{ -\frac{\tau}{2}\big[(n-1)s^2 + n(\mu - \bar{x})^2\big]\Big\},$$

$$\pi(\mu \mid \tau) \propto \tau^{1/2} e^{-\frac{\tau\kappa_0}{2}(\mu-\mu_0)^2}, \qquad \pi(\tau) \propto \tau^{\nu_0/2-1} e^{-\frac{\tau}{2}\nu_0\sigma_0^2}.$$

Completing the square in μ,

$$n(\mu - \bar{x})^2 + \kappa_0(\mu - \mu_0)^2 = \kappa_n(\mu - \mu_n(\mathbf{x}))^2 + \frac{n\kappa_0}{n + \kappa_0}(\bar{x} - \mu_0)^2,$$

with $\frac{n\kappa_0}{n+\kappa_0} = (\kappa_0^{-1} + n^{-1})^{-1}$, so

$$\pi(\mu, \tau \mid \mathbf{x}) \propto \tau^{1/2} e^{-\frac{\tau\kappa_n}{2}(\mu-\mu_n(\mathbf{x}))^2} \cdot \tau^{\nu_n/2-1} e^{-\frac{\tau}{2}\nu_n\sigma_n^2(\mathbf{x})},$$

(b) Given $\mathbf{x}$ and σ^2, $Z = \frac{\mu - \mu_n(\mathbf{x})}{\sigma/\sqrt{\kappa_n}} \sim N(0,1)$, free of σ^2, so $Z \perp V = \nu_n\sigma_n^2(\mathbf{x})/\sigma^2 \sim \chi^2(\nu_n)$, and $Z/\sqrt{V/\nu_n} \sim t(\nu_n)$. ∎

Without conjugacy $f_{\mathbf{X}}(\mathbf{x})$ is often intractable and MCMC is used; we keep conjugate priors.

11.2 Bayesian inference

Summarize the posterior: **posterior mean**, **median** or **mode**, with posterior variance or quantiles. $I(\mathbf{x})$ with $P(\theta \in I(\mathbf{x}) \mid \mathbf{X} = \mathbf{x}) = 1 - \alpha$ is a **credible interval**, a probability statement about θ given the data.

Example 11.5 (Normal mean, known variance)

The normal posterior has equal mean, median and mode:

$$\hat{\theta} = \frac{n\sigma^{-2}\bar{x} + \sigma_0^{-2}\mu_0}{n\sigma^{-2} + \sigma_0^{-2}},$$

weighting $\bar{x}$ and μ_0 by data precision $n\sigma^{-2}$ and prior precision σ_0^{-2}; credible interval $\hat{\theta} \pm z_{\alpha/2}\sigma(\mathbf{x})$. E.g. $\sigma = 10$, $n = 25$, $\bar{x} = 104$, prior $N(100, 5^2)$: $\hat{\theta} = (26 + 4)/0.29 = 103.45$, $\sigma(\mathbf{x}) = 1.857$, 95% interval 103.45 ± 3.64.

Example 11.6 (Normal model, conjugate prior)

By Theorem 11.2(b), for $\nu_n > 1$ the posterior of μ is a shifted, scaled t: mean, median and mode $\hat{\mu} = \mu_n(\mathbf{x})$, interval $\hat{\mu} \pm t_{\alpha/2}(\nu_n)\sigma_n(\mathbf{x})/\sqrt{\kappa_n}$.
For σ^2, with $Y = \nu_n\sigma_n^2(\mathbf{x})/\sigma^2 \mid \mathbf{x} \sim \chi^2(\nu_n)$,

$$\hat{\sigma}^{2,\text{mean}} = \nu_n\sigma_n^2(\mathbf{x})E[Y^{-1}] = \frac{\nu_n}{\nu_n - 2}\sigma_n^2(\mathbf{x}), \qquad \hat{\sigma}^{2,\text{median}} = \frac{\nu_n}{\chi^2_{0.5}(\nu_n)}\sigma_n^2(\mathbf{x}),$$

and $\hat{\sigma}^{2,\text{mode}} = \frac{\nu_n}{\nu_n+2}\sigma_n^2(\mathbf{x})$; all factors $\to 1$. The credible interval is

$$\frac{\nu_n\sigma_n^2(\mathbf{x})}{\chi^2_{\alpha/2}(\nu_n)} \le \sigma^2 \le \frac{\nu_n\sigma_n^2(\mathbf{x})}{\chi^2_{1-\alpha/2}(\nu_n)}.$$

Example 11.7 (Binomial–beta)

Prior $\text{Beta}(\alpha_1, \alpha_2)$, posterior $\text{Beta}(t + \alpha_1, n - t + \alpha_2)$, mean

$$\hat{\theta}^{\text{mean}} = \frac{t + \alpha_1}{n + \alpha_1 + \alpha_2} = \frac{n}{n + \alpha_1 + \alpha_2} \cdot \frac{t}{n} + \frac{\alpha_1 + \alpha_2}{n + \alpha_1 + \alpha_2} \cdot \frac{\alpha_1}{\alpha_1 + \alpha_2},$$

a weighted average of t/n and the prior mean; mode $\frac{(t+\alpha_1-1)_+}{(t+\alpha_1-1)_+ + (n-t+\alpha_2-1)_+}$. The credible interval $[B_{1-\alpha/2}, B_{\alpha/2}]$ uses upper Beta quantiles, from F tables via $\text{Beta}(r_1, r_2) \overset{d}{=} \frac{(r_1/r_2)F}{1+(r_1/r_2)F}$, $F \sim F(2r_1, 2r_2)$; for large n, $\hat{\theta} \pm z_{\alpha/2}\sqrt{\hat{\theta}(1-\hat{\theta})/(n + \alpha_1 + \alpha_2)}$. E.g. $t = 12$, $n = 40$, uniform prior: $\hat{\theta}^{\text{mean}} = 13/42 = 0.310$, interval 0.310 ± 0.140.

Bayesian testing. Compare the **posterior odds**

$$\frac{P(\theta \in \Omega_0 \mid \mathbf{X} = \mathbf{x})}{P(\theta \in \Omega_1 \mid \mathbf{X} = \mathbf{x})} = \frac{P(\theta \in \Omega_0 \mid \mathbf{X} = \mathbf{x})}{1 - P(\theta \in \Omega_0 \mid \mathbf{X} = \mathbf{x})}.$$

Example 11.8 (Simple hypotheses)

For $H_0 : \theta = \theta_0$ vs $H_1 : \theta = \theta_1$ with prior probabilities π_0, π_1,

$$\frac{P(\theta = \theta_0 \mid \mathbf{x})}{P(\theta = \theta_1 \mid \mathbf{x})} = \frac{f(\mathbf{x} \mid \theta_0)\pi_0}{f(\mathbf{x} \mid \theta_1)\pi_1}.$$

Rejecting when the odds are below 1 is the Bayes test of Theorem 9.1; for $N(\mu, 1)$ the odds are $\exp\{n(\mu_0 - \mu_1)(\bar{x} - \frac{\mu_0+\mu_1}{2}) + \log(\pi_0/\pi_1)\}$.

Example 11.9 (One-sided hypotheses)

(a) Normal model, conjugate prior, $H_0 : \mu \le \mu_0$: by Theorem 11.2(b)

$$P(\mu \le \mu_0 \mid \mathbf{x}) = F_{t(\nu_n)}\Big(\frac{\mu_0 - \mu_n(\mathbf{x})}{\sigma_n(\mathbf{x})/\sqrt{\kappa_n}}\Big),$$

($F_{t(\nu)}$ the $t(\nu)$ cdf).
(b) Binomial–beta, $H_0 : \theta \le \theta_0$: $P(\theta \le \theta_0 \mid \mathbf{x})$ is the posterior Beta cdf at θ_0, or approximately $\Phi((\theta_0 - \hat{\theta})/\sqrt{\hat{\theta}(1-\hat{\theta})/(n + \alpha_1 + \alpha_2)})$.

11.3 Bayes and minimax estimators

Frequentist inference evaluates procedures over repeated samples, e.g. by MSE; as in Chapter 8, the Bayes and maximum MSE are free of θ:

$$r(\pi, \hat{\theta}) = \int_\Omega E[(\hat{\theta} - \theta)^2 \mid \theta]\pi(\theta)\, d\theta, \qquad \max_{\theta \in \Omega} E[(\hat{\theta} - \theta)^2 \mid \theta].$$

Definition (Bayes and minimax estimators)

(a) $\hat{\theta}^\pi$ is a **Bayes estimator** for the prior π if $r(\pi, \hat{\theta}^\pi) = \min_{\hat{\theta}} r(\pi, \hat{\theta})$.
(b) $\hat{\theta}^m$ is a **minimax estimator** if $\max_\theta E[(\hat{\theta}^m - \theta)^2 \mid \theta] = \min_{\hat{\theta}} \max_\theta E[(\hat{\theta} - \theta)^2 \mid \theta]$.

Theorem 11.3 (Bayes estimator under squared error)

The Bayes estimator minimizes the posterior expected squared error and is the posterior mean:

$$\hat{\theta}^\pi(\mathbf{x}) = \arg\min_a E[(a - \theta)^2 \mid \mathbf{X} = \mathbf{x}] = E(\theta \mid \mathbf{X} = \mathbf{x}).$$

Proof. By iterated expectation

$$r(\pi, \hat{\theta}) = E[E\{(\hat{\theta}(\mathbf{X}) - \theta)^2 \mid \mathbf{X}\}].$$

so pointwise minimization of the inner conditional expectation minimizes r; and $E[(a - \theta)^2 \mid \mathbf{x}] = \mathrm{Var}(\theta \mid \mathbf{x}) + (a - E[\theta \mid \mathbf{x}])^2$. ■

Likewise for a **loss function** $L(\theta, \hat{\eta})$: the **Bayes risk** $r(\pi, \hat{\eta}) = E[L(\theta, \hat{\eta}(\mathbf{X}))]$ is minimized by minimizing the **posterior expected loss** for each $\mathbf{x}$; minimax uses the **risk function** $R_{\hat{\eta}}(\theta) = E[L(\theta, \hat{\eta}) \mid \theta]$.

Example 11.10

(a) Binomial–beta with squared error: $\hat{\theta}^\pi = \frac{t + \alpha_1}{n + \alpha_1 + \alpha_2}$.
(b) Normal model, conjugate prior, $L = (\hat{\mu} - \mu)^2/\sigma^2$: $\hat{\mu}^\pi = E(\mu\sigma^{-2} \mid \mathbf{x})/E(\sigma^{-2} \mid \mathbf{x}) = \mu_n(\mathbf{x})$, since $E(\mu \mid \sigma^2, \mathbf{x}) = \mu_n(\mathbf{x})$ is free of σ^2.
(c) $L = (\hat{\sigma}^2/\sigma^2 - 1)^2$: $\hat{\sigma}^{2,\pi} = E(\sigma^{-2} \mid \mathbf{x})/E(\sigma^{-4} \mid \mathbf{x})$, and with $EY = \nu_n$, $EY^2 = 2\nu_n + \nu_n^2$,

$$\hat{\sigma}^{2,\pi} = \nu_n \sigma_n^2(\mathbf{x}) \frac{\nu_n}{2\nu_n + \nu_n^2} = \frac{\nu_n}{\nu_n + 2} \sigma_n^2(\mathbf{x}),$$

the posterior mode.

Theorem 11.4 (Bayes estimators with constant risk are minimax)

If an estimator $\hat{\eta}^*$ has (a) constant risk, $R_{\hat{\eta}^*}(\theta) = R_{\hat{\eta}^*}(\theta_0)$ for all θ, and (b) is the Bayes estimator for some prior probability distribution π, then it is minimax.

Proof. By (a), $\max_\theta R_{\hat{\eta}^*} = r(\pi, \hat{\eta}^*)$; by (b), for any $\hat{\eta}$,

$$r(\pi, \hat{\eta}^*) \le r(\pi, \hat{\eta}) = \int R_{\hat{\eta}}(\theta)\pi(\theta)\, d\theta \le \max_\theta R_{\hat{\eta}}(\theta).$$

■

Example 11.11 (Minimax estimation of a binomial proportion)

Bernoulli(θ), squared error, estimators $a\hat\theta + b$, $\hat\theta = t/n$:

$$E[(a\hat\theta + b - \theta)^2 \mid \theta] = \frac{a^2\theta(1-\theta)}{n} + \{b + (a-1)\theta\}^2,$$

the θ^2 and θ coefficients vanish when $(1-a)^2 = a^2/n$ and $a^2/n - 2b(1-a) = 0$:

$$a = \frac{\sqrt{n}}{\sqrt{n}+1}, \qquad b = \frac{1}{2(\sqrt{n}+1)}.$$

$\hat\theta^* = \frac{t+\sqrt{n}/2}{n+\sqrt{n}}$ has constant risk $\frac{1}{4(\sqrt{n}+1)^2}$ and is Bayes for Beta($\sqrt{n}/2, \sqrt{n}/2$) (Example 11.10(a)): minimax (Theorem 11.4). Its maximum risk is below $1/(4n)$, but $\hat\theta$ is better except near $\theta = 1/2$.

A minimax estimator may only be a limit of Bayes estimators:

Theorem 11.5 (Least favourable sequence of priors)

If for some estimator $\hat\eta^*$ there are priors π_k with Bayes estimators $\hat\eta^{\pi_k}$ such that

$$\lim_{k\to\infty} r(\pi_k, \hat\eta^{\pi_k}) \geq \max_\theta R_{\hat\eta^*}(\theta),$$

then $\hat\eta^*$ is minimax.

Proof. For any $\hat\eta$, $r(\pi_k, \hat\eta^{\pi_k}) \leq r(\pi_k, \hat\eta) \leq \max_\theta R_{\hat\eta}$; let $k \to \infty$. ■

Example 11.12 (The sample mean is minimax)

$N(\mu, \sigma^2)$, loss $(\hat\mu - \mu)^2/\sigma^2$: the conjugate Bayes estimator $\mu_n(\mathbf{x})$ has Bayes risk $1/\kappa_n = 1/(n + \kappa_0)$. With $\kappa_0 = 1/k$, $r(\pi_k, \hat\mu^{\pi_k}) \to 1/n$, the constant risk of $\bar X$; by Theorem 11.5 $\bar X$ is minimax.

11.4 Selected exercises

Motivation. Book exercises: two more conjugate families, a Bayes estimator under relative loss, and a weighted loss making $\bar X$ Bayes and minimax.

Gamma(α, β) has density $\theta^{\alpha-1}e^{-\theta/\beta}/\{\Gamma(\alpha)\beta^\alpha\}$ (Example 11.2).

Exercise 11.1 (Pareto data, gamma prior)

Let $X_1, \dots, X_n$ have density $f(x;\theta) = \theta x^{-\theta-1}$ for $x > 1$ (Pareto$(1,\theta)$, $\theta > 0$), and let $\theta \sim$ Gamma(α, β). Find the posterior distribution of θ.

Solution.

Step 1. With $t = \sum_i \log x_i > 0$, $L(\theta;\mathbf{x}) \propto \theta^n e^{-\theta t}$.

Step 2. By Theorem 11.1,

$$\pi(\theta \mid \mathbf{x}) \propto \theta^n e^{-\theta t} \cdot \theta^{\alpha-1} e^{-\theta/\beta} = \theta^{n+\alpha-1} e^{-\theta(t+\beta^{-1})},$$

so $\theta \mid \mathbf{x} \sim$ Gamma$\big(n+\alpha, (t+\beta^{-1})^{-1}\big)$ (Example 11.3: $\log X_i$ is exponential with rate θ). ■

Exercise 11.2 (Multinomial data, Dirichlet prior)

Let $\mathbf{X} = (X_1, \dots, X_k)' \sim \text{Multi}(n, (p_1, \dots, p_k)')$ with $\sum_j p_j = 1$, and let $\theta = (p_1, \dots, p_{k-1})'$ have the Dirichlet$(\alpha_1, \dots, \alpha_k)$ prior, whose density on $\{p_j > 0,\ \sum_{j<k} p_j < 1\}$ is

$$\pi(\theta) = \frac{\Gamma(\alpha_1 + \cdots + \alpha_k)}{\Gamma(\alpha_1) \cdots \Gamma(\alpha_k)} \prod_{j=1}^{k} p_j^{\alpha_j - 1}, \qquad p_k = 1 - \sum_{j<k} p_j.$$

Find the posterior distribution of θ.

Solution.

Step 1. $L(\theta; \mathbf{x}) \propto \prod_j p_j^{x_j}$, $p_k = 1 - \sum_{j<k} p_j$.

Step 2. By Theorem 11.1, on the simplex,

$$\pi(\theta \mid \mathbf{x}) \propto \prod_{j=1}^{k} p_j^{x_j} \prod_{j=1}^{k} p_j^{\alpha_j - 1} = \prod_{j=1}^{k} p_j^{x_j + \alpha_j - 1},$$

the Dirichlet$(x_1 + \alpha_1, \dots, x_k + \alpha_k)$ kernel; so this is the posterior. For $k = 2$ it is Example 11.1; α_j act as prior counts. ■

Exercise 11.10 (A Bayes estimator under relative squared error)

In Exercise 11.1, find the Bayes estimator of θ for the loss $L(\theta, \hat{\theta}) = (\hat{\theta}/\theta - 1)^2$, assuming $n + \alpha > 2$.

Solution.

Step 1. Minimize the posterior expected loss (after Theorem 11.3):

$$E\left[(a/\theta - 1)^2 \mid \mathbf{x}\right] = a^2 E(\theta^{-2} \mid \mathbf{x}) - 2aE(\theta^{-1} \mid \mathbf{x}) + 1,$$

minimized at $a^* = E(\theta^{-1} \mid \mathbf{x}) / E(\theta^{-2} \mid \mathbf{x})$.

Step 2. For $\theta \sim \text{Gamma}(a_1, b)$, $k < a_1$,

$$E\theta^{-k} = \frac{\int_0^\infty \theta^{a_1 - k - 1} e^{-\theta/b}\, d\theta}{\Gamma(a_1) b^{a_1}} = \frac{\Gamma(a_1 - k)}{\Gamma(a_1)}\, b^{-k}.$$

so $E\theta^{-1} = \{(a_1 - 1)b\}^{-1}$, $E\theta^{-2} = \{(a_1 - 1)(a_1 - 2)b^2\}^{-1}$ $(a_1 > 2)$.

Step 3. With $a_1 = n + \alpha$, $b = (t + \beta^{-1})^{-1}$ (Exercise 11.1),

$$\hat{\theta}^{\pi} = (a_1 - 2)b = \frac{n + \alpha - 2}{\sum_i \log x_i + \beta^{-1}}.$$

smaller than the posterior mean $(n + \alpha)b$: relative loss penalizes overestimating small θ. ■

Exercise 11.12 (A weighted loss for a proportion)

Let $X_1, \dots, X_n$ be Bernoulli(θ), $0 < \theta < 1$, with loss $L(\theta, \hat{\theta}) = (\hat{\theta} - \theta)^2 / \{\theta(1 - \theta)\}$.
(a) Find the Bayes estimator for the uniform prior $U(0, 1)$.
(b) Find a minimax estimator of θ.

Solution.

(a)

Step 1. The posterior is $\text{Beta}(t+1, n-t+1)$ (Example 11.1), so the posterior expected loss of a is $\propto$

$$Q(a) = \int_0^1 (a-\theta)^2 \theta^{t-1}(1-\theta)^{n-t-1}\, d\theta.$$

Step 2. For $0 < t < n$, $Q(a) = a^2 B(t, n-t) - 2aB(t+1, n-t) + B(t+2, n-t)$ is minimized at

$$a^* = \frac{B(t+1,n-t)}{B(t,n-t)} = \frac{\Gamma(t+1)\Gamma(n)}{\Gamma(t)\Gamma(n+1)} = \frac{t}{n}.$$

Step 3. For $t = 0$, the integrand is $\sim a^2\theta^{-1}$ near 0, so $Q(a) = \infty$ unless $a = 0$, and $Q(0) < \infty$: $a^* = 0 = t/n$; symmetrically $a^* = 1$ for $t = n$.

Step 4. So $\hat{\theta}^\pi = \bar{X}$.

(b) $\bar{X}$ has risk

$$E\Big[\frac{(\bar{X}-\theta)^2}{\theta(1-\theta)} \;\Big|\; \theta\Big] = \frac{\text{Var}(\bar{X} \mid \theta)}{\theta(1-\theta)} = \frac{\theta(1-\theta)/n}{\theta(1-\theta)} = \frac{1}{n},$$

constant; with (a) and Theorem 11.4, $\bar{X}$ is minimax. (Under unweighted squared error it is not, Example 11.11; the weight equalizes its risk.) ■